(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] MARCH 2023

Paper Code: ETME-105 ***Subject: Manufacturing Processes***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Write **short notes** on the followings: - **(2.5x10=25)**

a) What is the difference between core print and chaplet?
b) Give the classification of steel based on carbon percentage.
c) Define ductility and brittleness.
d) What are the characteristics of moulding sand?
e) Function of the gating system.
f) Briefly explain the forging defects.
g) Briefly explain the Extrusion process with its application.
h) Concept of polarity in welding.
i) What is the use of Flux in welding?
j) Differentiate between coining and embossing?

UNIT-I

Q2 Explain the construction and working of the cupola furnace with neat diagram. **(12.5)**

Q3 a) Discuss various heat treatment processes. **(6)**
b) Explain the investment casting process with neat diagram. **(6.5)**

UNIT-II

Q4 Differentiate between hot working and cold working. Explain the types of forging operations **(12.5)**

Q5 Describe with the help of neat diagram explain the marking, measuring, holding and cutting tools used in the fitting shop **(12.5)**

UNIT-III

Q6 a) Explain the principle of electric arc welding with a neat diagram. **(6)**
b) What is the difference between TIG and MIG welding **(6.5)**

Q 7 a) Explain the Submerged arc welding with a neat diagram. **(6)**
b) Differentiate between welding, soldering and brazing. **(6.5)**

UNIT-IV

Q8 What is sheet metal, and what are the different materials used in sheet metal with its application? Explain the sheet metal operations. **(12.5)**

i) Shearing
ii) Drawing
iii) Nibbling
iv) Punching and blanking

Q9 Describe the powder metallurgy process. State the advantage and applications of powder metallurgy. **(12.5)**

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